FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Anatomy of the Genital Tract & Fetal Skull

Lecture | Week 1 · Foundations, ANC & early pregnancy | Foundations | Dr Ndiwalana

Labour is a mechanical problem of a passenger (the fetal skull) negotiating a passage (the bony pelvis and pelvic floor), driven by powers (the uterus). Knowing the pelvic diameters, the fetal skull landmarks and diameters, the supports and blood supply of the uterus, and the course of the ureter lets you diagnose position and CPD on vaginal examination, perform episiotomy, pudendal block and CS safely, and control haemorrhage.

Must know before the ward round

1. The pelvic inlet is widest transversely (about 13 cm) and the outlet is widest anteroposteriorly (about 13 cm), so the head enters transverse and rotates to occipito-anterior. The interspinous diameter (about 10 cm) is the narrowest and is the reference for station (0).
2. Obstetric conjugate (about 10.5–11 cm) ≈ diagonal conjugate (about 12.5 cm) minus 1.5 cm. If you can reach the sacral promontory on VE, the inlet is contracted.
3. A well-flexed vertex presents the suboccipitobregmatic diameter (9.5 cm). Deflexed or OP gives occipitofrontal (11.5 cm), brow gives mentovertical (13–13.5 cm, cannot deliver vaginally) and face gives submentobregmatic (9.5 cm).
4. Anterior fontanelle (bregma) is diamond-shaped with 4 sutures and closes at about 18 months. Posterior fontanelle (lambda) is Y-shaped with 3 sutures and closes at about 6–8 weeks. On VE, the posterior fontanelle near the symphysis means OA.
5. The uterine artery (from the anterior division of the internal iliac) crosses ABOVE the ureter about 1.5–2 cm lateral to the cervix ('water under the bridge'), which is the commonest site of ureteric injury at hysterectomy and CS.
6. The main uterine supports are the cardinal (transverse cervical/Mackenrodt's) and uterosacral ligaments plus levator ani. The round and broad ligaments are not significant supports.
7. The pudendal nerve (S2–S4) passes behind the ischial spine and supplies the perineum. Uterine pain in the first stage travels via T10–L1.
8. The ovarian artery comes from the aorta (L2). The right ovarian vein drains into the IVC and the left into the left renal vein. Ovarian lymph drains to the para-aortic nodes.
9. Engagement means the biparietal diameter (9.5 cm) has passed the brim, which is 2/5 or less of the head palpable abdominally. Abdominal fifths cannot be fooled by caput, so believe the abdomen over the VE station.
10. Moulding is graded 0 to +3. Moulding +3 (bones overlapping and irreducible) with a high head (3/5 or more palpable) means CPD or obstruction, so do a CS, not oxytocin or ventouse.
11. Clinical pelvimetry is adequate if the sacral promontory is not reached, the sacrum is hollow, the spines are not prominent, the subpubic angle admits 2 fingers and the intertuberous diameter admits 4 knuckles.
12. The best evidence of an adequate pelvis is a previous vaginal delivery of a good-sized baby. X-ray pelvimetry is not recommended routinely.

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1. Definitions and Key Concepts

Obstetrics is applied anatomy. Every labour is a meeting between the **passenger** (the fetal head: its diameters, sutures, fontanelles and ability to mould) and the **passage** (the bony pelvis and soft tissues), driven by the **powers** (the uterus). When you do a vaginal examination in labour you are reading anatomy with your fingers: which suture, which fontanelle, how much moulding, where the ischial spines are.

The one thing never to forget: **the head enters the pelvis transverse, rotates at the level of the spines, and must present its smallest diameter (9.5 cm, a well-flexed head) to get through.** Deflexion or a brow means trouble.

> CLINICAL PEARL
> **Passage, passenger, powers.** Every question about labour can be asked in this order: is the **passage** (bony pelvis, pelvic floor, cervix) adequate; is the **passenger** (size, attitude, position, moulding of the head) presenting its smallest diameter; and are the **powers** (contractions) adequate? Anatomy answers the first two.

Term	Meaning
True pelvis	The bony canal below the pelvic brim (linea terminalis). It is the birth canal
False pelvis	The part above the brim, between the iliac blades. It supports the gravid uterus but has no obstetric importance
Pelvic inlet (brim)	Bounded by the sacral promontory, alae of the sacrum, arcuate/iliopectineal lines, pectineal lines and upper border of the symphysis
Mid-pelvis (plane of least dimensions)	At the level of the **ischial spines**. It is the narrowest part
Pelvic outlet	A diamond made of two triangles: the anterior (pubic arch and ischial tuberosities) and posterior (tuberosities, sacrotuberous ligaments and tip of the coccyx/sacrum)
Station	Level of the lowest bony part of the fetal head relative to the **ischial spines** (0 = at the spines; −1 to −3 above; +1 to +3 below, in cm)
Engagement	The **widest transverse diameter** (biparietal, 9.5 cm) has passed through the pelvic brim. Abdominally, the head is **2/5 or less palpable**; vaginally, the leading part is at or near the spines
Moulding	Overlapping of the skull bones at the sutures under pressure. Graded 0 to +3
Caput succedaneum	Oedema of the scalp over the presenting part. It crosses suture lines
Attitude	Relationship of fetal parts to each other (flexion or deflexion of the head)
Denominator	The part of the presentation used to name position: **occiput** (vertex), **sacrum** (breech), **mentum** (face), **acromion** (shoulder)

2. Burden and Uganda Context

In Uganda, where **obstructed labour** still causes uterine rupture, fistula and maternal death, and where many CS and repairs are done by junior doctors at HC IVs, this anatomy is practical, not academic. It tells you **when a head will not fit**, **where the ureter is** when you clamp the uterine artery, and **where to inject** a pudendal block.

- **Obstructed labour and CPD** remain major causes of maternal death, **uterine rupture** and **obstetric fistula** in Uganda. Contributing factors are adolescent pregnancy (an immature pelvis), short stature from childhood undernutrition, late referral and unsupervised home births.
- **Obstetric fistula**: Uganda has an estimated tens of thousands of women living with unrepaired fistula. Fistula follows prolonged pressure of the fetal head on the bladder or rectum against the pubic bones, which causes ischaemic necrosis. This is anatomy at its most tragic.
- **Rickets** and pelvic fractures (road traffic accidents, boda boda injuries) cause occasional contracted pelvis.
- **Female genital mutilation** is practised in a few communities in eastern and north-eastern Uganda (e.g. among the Sebei and Pokot) and scars the vulva and perineum.
- Many Ugandan HC IVs rely on medical officers to perform CS, so **safe surgical anatomy** (lower segment, bladder, uterine vessels, ureters) is a core competence.

3. Principles and Framework

This section is the anatomy itself, organised from the passage (bony pelvis, pelvic floor, soft tissues) to the organs and their vessels and nerves, and finally the passenger (fetal skull).

1. The bony pelvis

Bones: two innominate (hip) bones (ilium, ischium and pubis fused at the acetabulum), the **sacrum** (5 fused vertebrae) and the **coccyx**. **Joints:** the symphysis pubis, two sacroiliac joints and the sacrococcygeal joint. These all relax slightly in pregnancy (relaxin), and the coccyx moves backwards in the second stage.

Obstetric diameters (average female pelvis):

Plane	Anteroposterior	Oblique	Transverse	Widest
Inlet (brim)	**11 cm** (obstetric/true conjugate)	12 cm	**13 cm**	**Transverse**
Mid-cavity	12 cm	12 cm	12 cm	Round
Mid-pelvis (spines)	about 11.5–12 cm	–	**Interspinous about 10 cm**	Narrowest part
Outlet	**13 cm** (with the coccyx pushed back)	12 cm	**11 cm** (intertuberous)	**Anteroposterior**

The conjugates of the inlet:

Conjugate	From	To	Length	Notes
Anatomical (true) conjugate	Sacral promontory	Upper border of the symphysis	about 11 cm	–
Obstetric conjugate	Sacral promontory	**Nearest point on the posterior surface of the symphysis**	about **10.5–11 cm**	The **smallest AP diameter the head must pass**; cannot be measured clinically
Diagonal conjugate	Sacral promontory	**Lower border** of the symphysis	about **12.5 cm**	**Measured on VE.** Obstetric conjugate ≈ diagonal conjugate − 1.5 cm

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            INLET viewed from above (gynaecoid)
                  Symphysis (front)
                 _____________________
               /                       \
              /     AP (true conj.) 11   \
             |  <------ transverse 13 --> |
              \          ^              /
               \_________|_____________/
                   Sacral promontory (back)

  The inlet is widest side-to-side, so the head ENTERS TRANSVERSE (OT).
  The outlet is widest front-to-back, so the head EXITS OA (after rotation).
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Axis of the birth canal (curve of Carus): the canal is J-shaped. It runs **downwards and backwards** in the upper part, then curves **forwards** around the pubic symphysis. This is why you pull **downwards first, then forwards/upwards** during a vacuum or forceps delivery.

Pelvic types (Caldwell–Moloy):

Type	Frequency (approximate)	Inlet shape	Features	Labour implication
Gynaecoid	about 50%	Round	Wide subpubic angle (**about 90°** or more), blunt spines, wide sacrosciatic notch	Ideal
Android	about 20–30%	Heart-shaped	Narrow subpubic angle, prominent spines, narrow fore-pelvis	**Deep transverse arrest**, OP positions, outlet problems
Anthropoid	about 20–25% (commoner in African women in some studies)	Long AP oval	Long AP, narrow transverse	**Direct OP** delivery ("face to pubes") is possible
Platypelloid	about 3–5%	Flat, wide oval	Short AP	Head engages transverse and may fail to engage; **transverse arrest at the inlet**

2. The pelvic floor and perineum

- **Pelvic diaphragm: levator ani** (pubococcygeus, puborectalis, iliococcygeus) plus **coccygeus**. It supports the pelvic organs, and in labour its gutter-shaped slope **encourages the leading part to rotate anteriorly** (the occiput to the front).
- **Nerve supply:** the **pudendal nerve (S2–S4)** and direct branches of S3–S4.
- **Urogenital diaphragm (perineal membrane)** with the superficial perineal muscles: bulbospongiosus, ischiocavernosus and superficial transverse perineal.
- **Perineal body (central tendon):** a fibromuscular node between the vagina and anus where bulbospongiosus, the superficial and deep transverse perineal muscles, the external anal sphincter and fibres of levator ani meet. It is **torn or cut in episiotomy** and must be carefully repaired to prevent prolapse and dyspareunia.
- **Anal sphincter complex:** the **external anal sphincter** (striated, voluntary) and **internal anal sphincter** (smooth, involuntary). Tears involving them are **3rd degree** (3a: less than 50% of the EAS; 3b: more than 50% of the EAS; 3c: EAS and IAS). A **4th degree** tear includes the anorectal mucosa.

Episiotomy: a **mediolateral** incision at **about 60° from the midline** at crowning (it avoids extension into the anal sphincter). It cuts the vaginal epithelium, perineal skin, bulbospongiosus, superficial transverse perineal muscle and sometimes fibres of levator ani.

3. The external genitalia (vulva)

Mons pubis, labia majora and minora, clitoris, vestibule (urethral and vaginal openings), **Bartholin's glands** (at 4 and 8 o'clock, where abscesses form), and the hymen. **Blood supply:** the internal pudendal artery (and external pudendal branches of the femoral artery). **Lymph drainage: superficial inguinal nodes**, which is why vulval cancer spreads to the groin.

4. The vagina

- A fibromuscular tube of about **7–8 cm on the anterior wall and 9–10 cm on the posterior wall**, directed upwards and backwards.
- The cervix projects into its vault, creating the **anterior, posterior and two lateral fornices**. The **posterior fornix is related to the pouch of Douglas** (rectouterine pouch), the lowest point of the peritoneal cavity, where blood and pus collect (culdocentesis, colpotomy).
- **Relations:** anteriorly the bladder and urethra (vesicovaginal and urethrovaginal fistula); posteriorly the pouch of Douglas, rectum and perineal body (rectovaginal fistula); laterally the ureters near the lateral fornices, and levator ani.
- **Epithelium:** non-keratinised stratified squamous with no glands. It is kept acidic (pH about 4.5) by **lactobacilli** (oestrogen-dependent glycogen).
- **Blood supply:** the vaginal artery (internal iliac) and branches of the uterine, middle rectal and internal pudendal arteries.
- **Lymph:** the upper third drains to the internal and external iliac nodes, the middle third to the internal iliac nodes and the lower third to the superficial inguinal nodes.

5. The uterus

- A pear-shaped, thick-walled muscular organ: about **7.5 × 5 × 2.5 cm**, **50–70 g** in the non-pregnant nulliparous woman.
- **Parts:** fundus, body (corpus), isthmus (the **lower uterine segment** in late pregnancy) and cervix (about 2.5–3 cm, with **internal and external os**). The body-to-cervix ratio is 2:1 in adults and 1:2 in prepubertal girls.
- **Position:** usually **anteverted** (the long axis of the cervix is tilted forwards relative to the vagina, about 90°) and **anteflexed** (the body is bent forwards on the cervix, about 170°). About 15–20% of women have a retroverted uterus.
- **Layers:** endometrium (functional and basal layers), **myometrium** (three layers; the **middle layer has interlacing "figure-of-eight" fibres around blood vessels, the "living ligatures"** that stop bleeding after placental separation), and perimetrium (peritoneum).
- **Peritoneal relations:** in front, the **uterovesical pouch** (opened at CS to push the bladder down). Behind, the **rectouterine pouch (of Douglas)**.

Lower uterine segment (LUS) vs upper segment:

Feature	Upper segment	Lower segment
Origin	Body of the uterus	Isthmus (forms from about 28 weeks; fully formed in labour)
Muscle	Thick, active, **contracts and retracts**	Thin, passive, stretches
Peritoneum	Firmly attached	**Loosely attached** (so the bladder is easy to reflect)
Clinical	Classical CS incision; bleeds more, heals worse, higher rupture risk (about 4–9%)	**Transverse lower segment CS**: less bleeding, stronger scar (rupture risk about 0.5–1%)

> **CLINICAL PEARL**
> In **obstructed labour**, the upper segment keeps retracting and the lower segment thins out. The junction rises as a visible groove, the **Bandl's ring (pathological retraction ring)**, which is a sign of **impending uterine rupture**. Do not confuse it with a full bladder.

6. Uterine supports

Structure	Anatomy	Importance
Cardinal (transverse cervical, Mackenrodt's) ligaments	From the cervix and upper vagina to the lateral pelvic wall. Contain the uterine vessels	**Principal support** of the uterus
Uterosacral ligaments	From the posterior cervix, around the rectum, to the sacrum	Hold the cervix back (keep the uterus anteverted). Main support
Pubocervical ligaments/fascia	Cervix to the pubis beneath the bladder	Support; damage causes cystocele
Levator ani / pelvic floor	See above	**Dynamic support**. Damaged in childbirth, causing prolapse
Round ligaments	Uterine cornu, through the inguinal canal, to the labia majora	Help keep anteversion. **Not a true support**. Stretching causes "round ligament pain" in pregnancy
Broad ligaments	Double fold of peritoneum from the uterus to the lateral pelvic wall	Contain the tubes, round and ovarian ligaments, vessels and ureter at the base. **Not a support**. Site of broad ligament haematoma

7. Fallopian tubes

About **10 cm** long, running in the upper free border of the broad ligament. The parts from medial to lateral are:

1. **Interstitial (intramural)**: within the uterine wall. An ectopic here ruptures late and bleeds heavily.
2. **Isthmus**: narrow. An ectopic here ruptures early. It is the site of tubal ligation.
3. **Ampulla**: the widest and longest part, the **site of fertilisation** and the **commonest site of ectopic pregnancy (about 70–80%)**.
4. **Infundibulum with fimbriae**: opens into the peritoneal cavity (so PID spreads to the peritoneum, and hydrosalpinx forms when it is blocked).

8. The ovaries

- Almond-sized (about 3 × 2 × 1 cm), in the **ovarian fossa** on the lateral pelvic wall, attached to the back of the broad ligament by the **mesovarium**, to the uterus by the **ovarian ligament**, and to the pelvic wall by the **infundibulopelvic (suspensory) ligament**, which carries the ovarian vessels.
- **Relations:** the obturator nerve lies in the lateral pelvic wall behind the ovary, so **ovarian pathology can refer pain to the medial thigh**. The ureter runs behind the ovary.
- **Blood supply:** the **ovarian artery from the abdominal aorta at L2**. Venous drainage: the **right ovarian vein to the IVC** and the **left ovarian vein to the left renal vein** (the same left–right asymmetry as the testicular veins in men).
- **Lymph: para-aortic nodes** at L2 (following the vessels), not the pelvic nodes.

9. Blood supply of the pelvis: the essentials

- **Internal iliac artery, anterior division: uterine artery**, vaginal, obturator, internal pudendal, inferior gluteal, umbilical/superior vesical, and middle rectal arteries.

- **Uterine artery:** runs medially in the base of the broad ligament (within the cardinal ligament), **crosses ABOVE (anterior to) the ureter about 1.5–2 cm lateral to the cervix at the level of the internal os**, then ascends tortuously along the side of the uterus to **anastomose with the ovarian artery**. Its branches are the **arcuate, radial and spiral arteries**. The spiral arteries are remodelled by trophoblast in pregnancy; this remodelling fails in pre-eclampsia.
- **Internal iliac artery ligation** (anterior division), done for intractable PPH, reduces pulse pressure (it does not fully stop flow, because of collaterals).

> RED FLAG
> **"Water under the bridge"**: the ureter (water) passes **under** the uterine artery (bridge) just lateral to the cervix. When clamping the uterine vessels at CS hysterectomy, or placing haemostatic sutures for lateral extensions of a CS incision, **stay close to the uterus and push the bladder down** to avoid ureteric ligation or injury. Also watch the ureter at the **pelvic brim** (crossing the bifurcation of the common iliac artery) and near the **infundibulopelvic ligament** when clamping ovarian vessels.

10. Nerve supply and pain pathways of labour

Stage	Pain source	Nerves	Analgesia target
First stage	Uterine contractions and cervical dilatation	Visceral afferents with sympathetic fibres to **T10–L1**	Epidural or spinal (T10–L1), paracervical block
Second stage	Stretching of the vagina, perineum and pelvic floor	**Pudendal nerve (S2–S4)**	Pudendal block, perineal infiltration, epidural/spinal extending to the sacral segments

Pudendal nerve course: it leaves the pelvis through the **greater sciatic foramen**, passes **behind the ischial spine and sacrospinous ligament**, re-enters through the **lesser sciatic foramen** into the **pudendal (Alcock's) canal**, and gives the inferior rectal nerve, perineal nerve and dorsal nerve of the clitoris. For a **transvaginal pudendal block**, palpate the ischial spine and inject just **medial and posterior to the tip of the spine**, through the sacrospinous ligament, after aspirating (the internal pudendal vessels are adjacent).

11. Lymphatic drainage: a summary for oncology

Organ	Drains to
Vulva and lower third of the vagina	**Superficial inguinal** nodes, then deep inguinal and external iliac
Cervix and upper vagina	**Parametrial, obturator, internal and external iliac**, then common iliac and para-aortic
Uterine body	Iliac and **para-aortic** nodes (the fundus drains along the ovarian vessels; some drainage via the round ligament to the inguinal nodes)
Ovary and tube	**Para-aortic** nodes

12. The fetal skull

Bones of the vault: two **frontal**, two **parietal**, two temporal (squamous parts) and one **occipital** bone. At birth they are not fused, which allows **moulding**.

Sutures:

Suture	Between	Clinical use
Sagittal	The two parietal bones	**The main landmark on VE**: its direction shows whether the head is transverse, oblique or AP

Suture	Between	Clinical use
Frontal (metopic)	The two frontal bones	Leads to the anterior fontanelle
Coronal	Frontal and parietal bones	Forms the lateral arms of the anterior fontanelle
Lambdoid	Parietal and occipital bones	Forms the arms of the posterior fontanelle

Fontanelles:

Feature	Anterior fontanelle (bregma)	Posterior fontanelle (lambda)
Shape	**Diamond (kite)**	**Triangle ("Y")**
Number of sutures	**4** (frontal, sagittal, 2 coronal)	**3** (sagittal, 2 lambdoid)
Size	Large (about 3 × 2 cm)	Small
Closes	About **18 months**	About **6–8 weeks**
On VE	Felt when the head is **deflexed** (OP, brow)	Felt when the head is **well flexed**

Regions of the skull: vertex (the area between the two fontanelles and the parietal eminences), **sinciput/brow** (in front of the anterior fontanelle, up to the root of the nose), **face** (root of the nose to the chin), **occiput** (behind the posterior fontanelle).

Fetal skull diameters: the heart of this lecture

Diameter	From → to	Length	Presentation
Suboccipitobregmatic	Below the occipital protuberance → centre of the anterior fontanelle	**9.5 cm**	**Vertex, well flexed** (normal)
Suboccipitofrontal	Below the occipital protuberance → centre of the forehead (sinciput)	**10 cm**	Vertex, partially flexed
Occipitofrontal	Root of the nose (glabella) → most prominent point of the occiput	**11.5 cm**	**Deflexed vertex / OP**
Mentovertical	Point of the chin → highest point of the vertex	**13–13.5 cm** (some books give 14)	**Brow**: the largest; **no vaginal delivery at term**
Submentovertical	Junction of the chin and neck → highest point of the vertex	**11.5 cm**	Face, partially extended
Submentobregmatic	Junction of the chin and neck → centre of the anterior fontanelle	**9.5 cm**	**Face, fully extended** (mento-anterior can deliver)

Transverse diameter	Between	Length
Biparietal	The two parietal eminences	**9.5 cm** (the widest transverse diameter; engagement means it has passed the brim)
Bitemporal	The furthest points of the coronal suture	**8 cm**
Bimastoid	Tips of the mastoid processes	7.5 cm

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ATTITUDE vs PRESENTING DIAMETER

Well flexed vertex  ── suboccipitobregmatic ── 9.5 cm     easy
Partly flexed       ── suboccipitofrontal ──── 10  cm
Deflexed / OP       ── occipitofrontal ─────── 11.5 cm    (long labour)
Brow                ── mentovertical ──────── 13–13.5 cm obstructs
Face (extended)     ── submentobregmatic ──── 9.5 cm     delivers if mento-anterior

(The more the head extends, the bigger the diameter, until full
 extension (face) makes it small again.)
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Moulding and caput:

- **Moulding** means the parietal bones overlap the occipital and frontal bones, and one parietal bone overlaps the other at the sagittal suture. It **reduces the engaging diameter by up to about 1 cm**. Grades on the partograph:

Grade	Finding
0	Bones separated; sutures easily felt
+1	Bones just touching
+2	Bones overlapping but can be separated with finger pressure
+3	Bones overlapping and **cannot be separated**, which suggests **CPD**

- **Caput succedaneum**: diffuse oedema of the scalp, **crosses suture lines**, present at birth, resolves in days. Large caput plus +3 moulding in a poorly progressing labour means **obstruction**.
- **Cephalhaematoma**: subperiosteal bleed, **does not cross sutures** (limited by the periosteum), appears hours after birth, resolves over weeks, and can cause jaundice.
- **Subgaleal haemorrhage**: a bleed under the aponeurosis that crosses sutures and can cause **hypovolaemic shock** in the newborn. It is associated with vacuum delivery.

Intracranial membranes: the falx cerebri and tentorium cerebelli. Excessive or abnormal moulding (especially rapid, as in precipitate breech delivery of the aftercoming head) can **tear the tentorium**, rupturing the great cerebral vein (of Galen) and causing intracranial haemorrhage.

13. Anatomy predicts the mechanism of labour

The head meets each plane of the pelvis with the diameter that fits it best. Each cardinal movement is a direct consequence of the anatomy above.

Inlet widest transversely: head engages OT, biparietal 9.5 cm passes the brim

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Flexion: suboccipitobregmatic 9.5 cm presents

↓

Descent through the round mid-cavity

↓

Internal rotation at the spines: levator ani gutter turns the occiput forwards

↓

Outlet widest AP: head arrives OA

↓

Extension under the symphysis along the curve of Carus

↓

Restitution and external rotation as the shoulders rotate

↓

Delivery of the shoulders and body

4. Application in Practice

The applied workup of this topic is the **"passage and passenger" assessment**: can this head get through this pelvis? You do it on every woman admitted in labour, again whenever progress is slow, and always before an instrumental delivery or augmentation. It is also the workup that tells you **when to stop and do a caesarean section** before the uterus ruptures or a fistula forms.

Step 0: First 5 minutes

On arrival in labour (or when called to a "slow labour"):

1. **Vital signs:** pulse, BP, temperature, RR. A woman in **obstructed labour** is often **tachycardic, dehydrated (dry tongue, ketones), febrile** and exhausted.
2. **Fetal heart** with Doppler or Pinard (110–160 bpm), listening through and after a contraction.
3. **Quick abdominal look and feel** for the emergencies that change everything:
   - **Bandl's ring**, a tense tender lower segment, a distended bladder → **obstructed labour**.
   - Sudden pain then contractions stop, fetal parts easily felt, shock → **uterine rupture**.
   - Transverse lie with ruptured membranes or a prolapsed arm or cord → emergency CS.
4. **Any vaginal bleeding before delivery? No digital VE** until placenta praevia is excluded (scan or examination in theatre).
5. **If obstruction or rupture is suspected: call for help now**, two large-bore cannulae, Hb and group and cross-match, IV fluids, catheter, **stop any oxytocin**, and prepare theatre.

Step 1: Focused history

Ask	Why: the answer that changes the plan
Parity and previous deliveries: vaginal? size of the biggest baby? instrumental?	**A previous vaginal delivery of a 3.5 kg or bigger baby is the best proof of an adequate pelvis.** A primigravida has an "untested pelvis"
Previous obstructed labour, stillbirth in labour, CS for "CPD" or "failure to progress", fistula	Suggests a small pelvis. Previous CS for CPD → usually a repeat CS (discuss with a senior)
Type of previous CS incision (classical or lower segment)	Classical scar → rupture risk too high for labour

Ask	Why: the answer that changes the plan
Childhood rickets, polio, TB spine, limp; pelvic or hip fracture (boda boda accidents)	Bony deformity → contracted or asymmetric pelvis
Age (adolescent) and **height**	Adolescent and under 150 cm → immature or small pelvis
Onset of labour, time of rupture of membranes, liquor colour, time pushing	Prolonged labour and long ROM → infection, fistula risk, moulding
Estimated gestation and size (GDM, big previous babies)	Macrosomia → relative CPD, shoulder dystocia
Previous 3rd/4th degree tear, FGM	Perineal plan (episiotomy, repair by an experienced person)
Leaking urine or faeces after a previous delivery	**Fistula** (vesicovaginal or rectovaginal) → plan repair and the mode of the next delivery with a senior

Step 2: Focused examination

Examine in the order of Hutchison's Clinical Methods (24th edition), adapted for the labour ward. In an emergency (Step 0) resuscitate first and complete the sequence once she is stable.

Introduction, consent and preparation

- **Introduce yourself**, confirm her name and age, and explain what you are going to do and why.
- **Consent** for the abdominal examination and, separately, for each vaginal examination; offer a **female chaperone**, use a screen and keep her covered except for the part being examined (respectful maternity care).
- **Hand hygiene** before touching her; sterile gloves and an antiseptic swab for the VE.
- **Bladder**: ask her to empty it first (a full bladder distorts the fundal height and the fifths).
- **Position and exposure**: semi-recumbent with a **left lateral tilt** for the abdomen (from about 20 weeks, to avoid aortocaval compression), exposed from the xiphisternum to the symphysis; dorsal or lithotomy position for the VE.

General inspection and vital signs

- **From the end of the bed**: well or unwell, exhausted, distressed, pushing; dehydration (dry tongue); pallor.
- **Build**: height (**below 150 cm** is a risk marker), gait and limp, spinal deformity, signs of rickets (bowed legs).
- **Vital signs**: pulse, BP, temperature, RR; a woman in obstructed labour is often **tachycardic, febrile and ketotic** (Step 0).

Hands, face and neck

- Hands: pallor, capillary refill, cool peripheries (shock in rupture or haemorrhage).
- Face and mouth: conjunctival pallor, dry tongue (dehydration in prolonged labour).
- Note anything that will matter in theatre (anaemia, difficult airway); then move to the abdomen.

Obstetric abdominal examination (the passenger from above)

In order:

1. **Inspection**: size and shape of the uterus, **scars** (previous CS: lower transverse or vertical/classical), a visible **Bandl's ring**, a distended bladder above the uterus.
2. **Fundal height and size**: symphysis–fundal height and **estimated fetal weight** (clinical estimate).
3. **Lie and presentation.**
4. **Engagement: fifths of the head palpable above the brim** (5/5 free; **2/5 or less = engaged**). This is the **most reliable measure of descent** because caput cannot fool it.

5. **Contractions**: frequency per 10 minutes and duration; palpate the bladder (distended?), a **Bandl's ring** and **tenderness of the lower segment**. A Bandl's ring, a distended bladder and a tense tender lower segment are signs of obstruction.
6. **Auscultation of the fetal heart** after a contraction (Pinard or Doppler, 110–160 bpm; Step 0).

Vaginal examination (read the head with your fingers)

The sequence:

1. Cervix: dilatation, effacement, application to the head.
2. Membranes: intact or ruptured; colour of the liquor.
3. **Presenting part:** vertex (smooth, sutures, fontanelles), face (mouth, nose, malar bones, gums) or breech (anus, sacrum, ischial tuberosities, genitalia).
4. **Position:** follow the **sagittal suture** to a fontanelle.
   - The **posterior fontanelle** (small, 3 sutures, Y) **anterior** means **OA** (good).
   - The **anterior fontanelle** (large, 4 sutures, diamond) **easily felt** means **deflexion** (OP or brow).
   - Sagittal suture **transverse** means OT (normal at the inlet; a **deep transverse arrest** if it is still transverse at or below the spines in the second stage).
5. **Station** relative to the ischial spines.
6. **Moulding** (0 to +3) and **caput**.
7. **Clinical pelvimetry** (table below).

Vaginal examination: the passenger (after hand hygiene, consent, and with a chaperone):

Feature	How to assess	Normal / reassuring	Worrying
Cervix	Dilatation, effacement, application	Well applied, dilating at about 1 cm/h in active labour	**Oedematous, loosely applied**, stalled dilatation
Presenting part	Vertex, face, brow, breech	Vertex	**Brow** (anterior fontanelle + supraorbital ridges + root of nose)
Position	Follow the sagittal suture to a fontanelle; **count its sutures (3 = posterior, 4 = anterior)**	OA (small Y-shaped posterior fontanelle anteriorly)	OP; sagittal suture still transverse at or below the spines in the second stage (deep transverse arrest)
Attitude	Which fontanelle is easiest to feel	Only the posterior fontanelle felt (well flexed, 9.5 cm)	Anterior fontanelle easily felt (deflexed, 11.5 cm)
Station	Bony leading part relative to the ischial spines	Descending with time (−2 → 0 → +2)	No descent; station "low" only because of **caput**
Moulding	Parietal bones at the sagittal suture; occipital bone at the lambdoid suture	0 or +1	**+3 (overlapping, irreducible)**
Caput	Scalp oedema over the presenting part	None or small	Large caput with a high head
Liquor	Colour and smell	Clear	Meconium, blood, offensive

Vaginal examination: the passage (clinical pelvimetry), best done at the first VE in labour or at 36 weeks:

Feature	Technique	Adequate	Suggests a contracted pelvis
Diagonal conjugate	Try to reach the sacral promontory with the middle finger	**Promontory not reached** (diagonal conjugate about 12.5 cm or more)	Promontory easily reached

Feature	Technique	Adequate	Suggests a contracted pelvis
Sacral curve	Run the fingers down the sacrum	Hollow, well curved	Flat or forward-projecting sacrum
Side walls	Sweep laterally	Parallel	Converging
Ischial spines	Feel both	Blunt, not prominent (interspinous about 10 cm or more)	**Prominent, close together** (android)
Sacrospinous ligament	Fingers from spine to sacrum	Admits **2 fingers or more**	Narrow
Subpubic angle	Two fingers under the pubic arch	Admits **2 fingers** (about 90°)	Narrow (below 2 fingers)
Intertuberous diameter	Closed fist between the ischial tuberosities (outside)	**4 knuckles** (about 8.5 cm or more)	Fewer than 4 knuckles
Coccyx	Press backwards	Mobile	Fixed, forward-angled

EXAM FAVOURITE

In the OSCE, describe pelvimetry in the order **inlet → cavity → mid-pelvis → outlet**: promontory, sacral curve and side walls, spines and sacrospinous ligament, then subpubic angle, intertuberous diameter and coccyx. Finish with "**clinically adequate pelvis**" or "**clinically contracted at the … level**".

Completing the examination

- Remove gloves, cover her, help her into a comfortable position (left lateral), **thank her** and explain the findings.
- **Hand hygiene**.
- **Plot** the findings on the partograph (dilatation, descent in fifths, moulding, liquor, FHR, contractions).
- **Summarise** in one sentence with an impression and a plan (model VE report in Step 5).

Step 3: Bedside tests

Test	What you look for	How it changes the plan
Partograph (start in active labour, cervix 4 cm or more on the modified WHO partograph)	Cervicograph against the alert and action lines; descent in fifths ("O"); **moulding** and liquor with each VE	Crossing the **action line** → full clinician assessment for CPD. **Dilatation stalling + no descent + moulding +3 = obstruction** → CS, not oxytocin
Doppler / Pinard FHR	110–160 bpm; decelerations after contractions	Fetal distress with obstruction → urgent CS
Urine (catheter or void)	**Ketones** (dehydration), **blood** (bladder compression in obstruction), volume	Haematuria in labour = bladder under pressure → obstruction until proven otherwise; catheterise
HemoCue Hb	Baseline before likely CS	Below 7 g/dL → blood ready before theatre
Bedside ultrasound (if trained)	Fetal spine and **orbits** show position (orbits anterior = OP); presentation; FHR	Confirms OP, face or brow when the VE is unclear because of caput

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management (and Ugandan availability)
Obstetric ultrasound	Uncertain presentation (brow, face, breech), high head at term in a primigravida, suspected macrosomia or placenta praevia	Presentation, position, estimated fetal weight, placental site, liquor	Brow at term or praevia → CS; macrosomia → senior plan. HC IV (often) and all hospitals
Intrapartum ultrasound for head position	Second stage, before instrumental delivery if position unclear	Occiput position (OA/OP/OT)	Wrong position is a common cause of failed ventouse. Mainly referral hospitals
Hb and group and cross-match	Obstructed labour, likely CS, previous CS	Hb; blood available	Anaemia plus likely haemorrhage → have 2 units ready. HC IV and hospitals
Full blood count, blood culture	Prolonged labour with fever	Raised WCC (interpret with labour norms), bacteraemia	Sepsis antibiotics (ampicillin + gentamicin ± metronidazole). Culture at regional/national referral
Urea and electrolytes, creatinine	Prolonged obstructed labour, dehydration, oliguria	Raised creatinine (above about 90 µmol/L in pregnancy), low potassium	Fluid resuscitation, catheter, watch renal function. Hospital labs
X-ray / CT pelvimetry	**Not routine** (no proven benefit); occasionally after a pelvic fracture or before planning birth with known deformity	Measured diameters	Rarely changes practice; labour itself is the best pelvimeter. CT at national referral/private centres
Dye test (methylene blue) / examination under anaesthesia	Leaking urine after obstructed labour	Fistula site and size	Continuous catheter for 10–14 days (small fresh fistula may heal); referral to a fistula repair centre (e.g. Mulago, Kitovu Hospital, or regional fistula camps)

The investigations for this topic are mostly **clinical**: abdominal fifths, VE with clinical pelvimetry, and the partograph. They are set out, with their interpretation and Ugandan availability, in **Steps 2–4** above. Remember that **X-ray or CT pelvimetry is not recommended routinely** (it does not improve outcomes), and that **labour itself is the best test of the pelvis**. After obstructed labour, a **methylene blue dye test** (with cystoscopy at a fistula centre) assesses a suspected vesicovaginal fistula.

Step 5: Putting it together

Model one-line summary:

> "Ms A.T., 17-year-old primigravida, **height 146 cm**, at term, **in labour for 14 hours**, membranes ruptured 9 hours ago with clear liquor. Pulse 108, T 37.6°C, ketones 2+. **Head 3/5 palpable**, cervix **7 cm for the last 4 hours** (crossed the action line), **oedematous**, OT position, station –1, **moulding +3, caput ++**. Pelvimetry: promontory reached, spines prominent, subpubic angle 1.5 fingers. FHR 150 with no decelerations."

Problem list: 1) arrest of the active phase with signs of **cephalopelvic disproportion**; 2) contracted pelvis clinically (adolescent, short stature); 3) maternal dehydration and early infection risk (tachycardia, ROM 9 hours); 4) fistula and uterine rupture risk.

Working diagnosis: "Obstructed labour secondary to **CPD** in a primigravida with a clinically contracted pelvis." **Plan:** no oxytocin; IV fluids, catheter, antibiotics, Hb and cross-match; **emergency lower-segment CS**; keep the catheter for 10–14 days after delivery.

Worked example 1: reading a head that "looks low"

Finding	Reading
VE: "vertex at station +2"	Sounds as if the head is deep in the pelvis
But: **large caput ++**, **moulding +3**, sutures hard to feel	The fingertip is on **scalp oedema**, not bone
Abdomen: **head 3/5 palpable**	The widest diameter has **not** passed the brim, so the head is **not engaged**
Conclusion	The head is high and obstructed. **Not suitable for ventouse** (needs 1/5 or less palpable and bony station at or below the spines). Deliver by CS

Worked example 2: OP or deep transverse arrest, and what the anatomy tells you

A multipara has been fully dilated for 2 hours (pushing for 1 hour). The head is **0/5–1/5 palpable**, bony station **+1**, the **sagittal suture is transverse**, moulding +1, contractions 4 in 10 minutes.

- The head is **engaged and at or below the spines** with little moulding, so this is **not obstruction**. It is a **deep transverse arrest**: the head has failed to rotate at the mid-pelvis (often an android pelvis with prominent spines or weak pelvic floor rotation).
- Management: an experienced operator may try **ventouse** (with a posterior/OP cup placed at the flexion point, allowing autorotation) or manual rotation; otherwise CS. Compare this with example 1, where the head was **3/5 palpable with +3 moulding**: that is obstruction, and any vaginal attempt would be dangerous.

> **CLINICAL PEARL**
> The abdominal **fifths** tell you where the **bone** is; the VE station can be fooled by caput. When the two disagree, **believe the abdomen** and assume the head is higher than it feels.

Model VE report (labour ward)

> "On vaginal examination the vulva and vagina are normal. The cervix is **6 cm dilated, 80% effaced, soft and well applied** to the presenting part. Membranes are **ruptured with clear liquor**. The presenting part is the **vertex** at **station –1**. The **sagittal suture is in the right oblique diameter** with the **posterior fontanelle anterior on the left**, so **left occipito-anterior (LOA)**. **Moulding +1, no caput.** The pelvis feels **adequate**: sacral promontory not reached, spines not prominent, subpubic angle admits 2 fingers. **Impression: normal progress in the active first stage.**"

5. Clinical Acumen

Thinking like a clinician

- **Pelvis + head + powers:** before saying "CPD", ask whether the **powers** are adequate (augment if not, provided there is no obstruction), whether the **position** is right (OP and deflexion enlarge the diameter), and only then whether the passage is too small.
- **Anatomy predicts the mechanism:** the head enters **transverse** (widest inlet diameter), **flexes** (so 9.5 cm presents), **descends**, **rotates internally** at the spines/pelvic floor to OA (the outlet is widest AP), **extends** under the symphysis, **restitutes**, and **rotates externally**, then the shoulders are delivered.
- **Previous vaginal delivery of a 3.5 kg baby** is better evidence of an adequate pelvis than any X-ray pelvimetry (which is not recommended routinely).
- **Pitfall:** calling OP "OA" because the fontanelles were confused. Count the sutures meeting at the fontanelle (**3 = posterior**, **4 = anterior**), and feel for the size.

- **Pitfall:** pushing a brow presentation through with oxytocin. It will not deliver: 13 cm will not pass a 10–11 cm pelvis.
- **Surgery:** at CS, reflect the **bladder** down from the lower segment and make the incision **transverse in the LUS**. For lateral extensions, **place the fingers under the broad ligament and identify the ureter**.

Red flags

RED FLAG

Signs of obstructed labour / CPD, so call a senior and prepare for CS:

- Head still **3/5 or more palpable** abdominally with a fully dilated cervix
- **Moulding +3** and large caput
- **Cervix oedematous**, poorly applied, dilatation stalls (crosses the action line on the partograph)
- **Bandl's ring**, haematuria (bladder compression), maternal tachycardia, dehydration, fever
- A **brow presentation** (anterior fontanelle, supraorbital ridges and nose felt) at term
- Fetal distress

Beware: a large caput may reach the spines while the bony head is still high, giving a **false impression of descent**. Always correlate the VE with the **abdominal fifths**.

Scenarios: what is the presenting part on VE?

Finding on VE	Distinguishing features	What it means
Vertex, OA	Sagittal suture in the oblique/AP diameter; the **small Y-shaped posterior fontanelle** anteriorly	Normal
Vertex, OP	**Diamond anterior fontanelle** felt anteriorly; posterior fontanelle posteriorly and hard to reach	Longer labour; backache; may rotate
Vertex, deep transverse arrest	Sagittal suture **transverse** at or below the spines in the second stage, no progress	Android pelvis, weak powers; needs rotation (ventouse, manual) or CS
Brow	Anterior fontanelle, **supraorbital ridges, root of the nose**; mentovertical 13–13.5 cm	CS at term unless it converts
Face	**Mouth with gums, nose, malar eminences** (a triangle); no fontanelles	Mento-anterior can deliver; mento-posterior means CS
Breech	**Anus in line with the ischial tuberosities** (a line, not a triangle); meconium on the glove; sacrum; genitalia	Breech management
Caput mistaken for descent	Soft, boggy swelling; sutures hard to feel; the abdomen still has 3/5 or more palpable	Obstruction

Common student mistakes

- Reporting station from the VE without checking the **abdominal fifths** (caput fools the fingers).
- Calling OP "OA" because the fontanelles were not identified by **counting the sutures**.
- Starting **oxytocin** for "poor progress" before excluding obstruction.
- Forgetting to **catheterise** after obstructed labour, or removing the catheter after 1–2 days.
- Not doing a **rectal examination** before and after perineal repair.
- Clamping lateral to the uterus at CS without pushing the bladder down and identifying the **ureter**.

6. Common Problems and Their Management

This lecture underpins management in labour and surgery. The practical applications follow.

1. Suspected CPD or obstructed labour at a HC IV: step by step

1. **Recognise** it: poor progress on the partograph plus +3 moulding, caput, and a high head (3/5 or more palpable).
2. **Call for help** (medical officer, theatre team). Do **not** augment with oxytocin if obstruction is suspected (risk of rupture).
3. **Resuscitate**: IV access (2 large-bore cannulae), send blood for **Hb and grouping and cross-match**, start **IV fluids** (e.g. Ringer's lactate 1 L over 1–2 hours if dehydrated), and **catheterise** the bladder (it relieves bladder compression, lets you measure urine output, and helps prevent fistula).
4. **Antibiotics** if labour is prolonged, the membranes are ruptured or there is fever: e.g. **ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily** (+ **metronidazole 500 mg IV 8-hourly** if chorioamnionitis is suspected), or prophylactic **cefazolin 2 g IV** at CS.
5. **Deliver by caesarean section** (lower segment). Expect a **deeply impacted head**: an assistant can **push the head up vaginally**, or use the **reverse breech extraction (Patwardhan)** technique.
6. **Symphysiotomy** or **destructive procedures** (craniotomy for a dead fetus) are historical or rare options at some units; follow local expertise.
7. **After delivery:** keep the **urinary catheter for 10–14 days** after prolonged obstructed labour (it prevents or may heal early fistula). Watch for PPH (atony), sepsis and uterine rupture signs.

2. Pudendal block (for outlet instrumental delivery or perineal repair)

1. Explain, obtain consent, position in lithotomy.
2. **Lidocaine 1%, up to 10 mL each side** (maximum lidocaine without adrenaline **3 mg/kg**, about 200 mg in total, so 20 mL of 1% is already at the limit; if you also need perineal infiltration, dilute to **0.5%** as in the WHO MCPC manual so the total stays within the maximum).
3. Palpate the **ischial spine** vaginally with the index and middle fingers. Guide a long needle (with a guard, e.g. an Iowa trumpet) to **just below and medial to the tip of the spine**, through the **sacrospinous ligament**.
4. **Aspirate** (the internal pudendal artery and vein lie next to the nerve), then inject about 10 mL.
5. Repeat on the other side. Wait 3–5 minutes and test sensation.
6. **Complications:** intravascular injection (seizures, arrhythmia), haematoma, infection.

3. Episiotomy and perineal repair: anatomy-driven steps

1. **Indications:** a rigid perineum delaying delivery, instrumental delivery, fetal distress in the second stage. **Not routine.**
2. **Infiltrate** with lidocaine 1% (about 10 mL) along the planned line.
3. Cut a **mediolateral** incision starting at the fourchette and directed **about 60° from the midline** at crowning, during a contraction, while protecting the fetal head.
4. **Repair** (after checking for a 3rd/4th degree tear with a **rectal examination**), using **absorbable polyglactin (Vicryl Rapide) 2-0**:
   - Vaginal epithelium: a continuous suture, **starting 1 cm above the apex**, to the hymenal ring.
   - **Perineal muscles** (bulbospongiosus, transverse perineal): continuous or interrupted sutures to rebuild the **perineal body**.
   - Skin: **subcuticular**.
5. **Rectal examination after repair** to make sure no suture has gone through the rectum. Count swabs and needles.

6. **Aftercare:** analgesia (paracetamol 1 g 6-hourly, or diclofenac 50 mg 8-hourly if not contraindicated), perineal hygiene (sitz baths), and laxatives if there was a 3rd/4th degree tear. For OASIS (3rd/4th degree tears), give broad-spectrum antibiotics (e.g. **cefuroxime + metronidazole**) and **repair in theatre by an experienced surgeon**.

4. Protecting the ureter and bladder at CS and hysterectomy

1. Open the **uterovesical peritoneum** and **reflect the bladder downwards** before incising the lower segment.
2. Make the uterine incision **transverse, curving slightly upwards laterally** to avoid the uterine vessels.
3. For **lateral extension with bleeding**: pack, call for help, and identify the ureter (palpate it at the pelvic brim or through the posterior leaf of the broad ligament). Place sutures **close to the uterus**.
4. **Uterine artery ligation (O'Leary)** for PPH: a stitch through the myometrium **2–3 cm medial** to the vessels at the level of the lower segment, which avoids the ureter.
5. At **hysterectomy**, clamp the **infundibulopelvic ligament** (if removing the ovaries) or the **ovarian ligament and tube** (if conserving them) close to the uterus, and the **uterine vessels at the level of the internal os, hugging the cervix**.
6. **Suspect injury** if there is haematuria, anuria or a flank mass after surgery, or leaking urine vaginally (fistula) days later.

5. Managing a newborn scalp swelling

First tell the swellings apart:

Condition	Crosses sutures?	Timing	Risk
Caput succedaneum	**Yes**	Present at birth, resolves in 1–3 days	Benign
Cephalhaematoma	**No** (limited to one bone)	Appears after hours; resolves over weeks	Jaundice; rarely an underlying fracture
Subgaleal haemorrhage	**Yes** (fluctuant, spreads to the neck and ears)	Hours after (often vacuum) delivery	**Hypovolaemic shock**. Monitor Hb, HR and head circumference
Chignon	Localised to the cup site	After vacuum	Benign; resolves in hours

Swelling	Management
Caput	Reassure; resolves in days
Cephalhaematoma	Do **not aspirate** (infection risk). Monitor for **jaundice** (phototherapy if needed) and anaemia; resolves in weeks
Subgaleal haemorrhage	**Emergency**: admit to the neonatal unit, monitor HR, BP and head circumference, give **vitamin K 1 mg IM** (if not given), **volume replacement and blood transfusion**, and correct coagulopathy

6. Emergency flow: uterine rupture suspected in obstructed labour

1. **Signs:** sudden severe pain, then cessation of contractions, maternal shock, fetal parts easily palpable, loss of station, fetal heart absent, vaginal bleeding, haematuria.
2. **Resuscitate:** call for help, 2 large-bore cannulae, **cross-match 2–4 units**, crystalloid, oxygen, catheter.
3. **Laparotomy** as soon as possible: deliver the baby and placenta, then **repair** the rupture (if it is simple, e.g. a lower segment tear, and she wants more children) or do a **subtotal/total hysterectomy** (if it is extensive or she is unstable). **Identify the ureters and bladder** because lateral tears extend into the broad ligament.
4. **Antibiotics**, tubal ligation if she consents (repair carries a high risk of repeat rupture), and a **catheter for 10–14 days** if the bladder was involved.

7. Complications of the anatomical problems

Anatomical problem	Complication
Contracted pelvis / CPD	Obstructed labour, uterine rupture, **VVF/RVF**, fetal asphyxia, intracranial haemorrhage, maternal death
Prolonged head compression on the pelvic floor	**Obstetric fistula**, foot drop (lumbosacral trunk or common peroneal nerve compression)
Perineal trauma	3rd/4th degree tears leading to **faecal incontinence**; dyspareunia; prolapse later
Damage to the pelvic floor and cardinal/uterosacral ligaments	**Uterovaginal prolapse** (especially with grand multiparity, heavy work and early resumption of labour postpartum)
Ureteric injury at surgery	Ureterovaginal fistula, hydronephrosis, loss of the kidney
Excessive or rapid moulding	Tentorial tear, intracranial haemorrhage
Pudendal block	Haematoma, local anaesthetic toxicity

Drug summary for this topic

All doses are as already given in the steps above.

Drug	Dose	Route	Frequency	Key cautions
Lidocaine 1% (pudendal block, episiotomy, perineal infiltration)	Up to 10 mL each side for a pudendal block; about 10 mL for episiotomy infiltration	Local infiltration	Once	Maximum **3 mg/kg without adrenaline** (about 200 mg); dilute to 0.5% if a larger volume is needed; **aspirate** before injecting
Ampicillin	2 g	IV	6-hourly	Prolonged labour, ruptured membranes, fever
Gentamicin	5 mg/kg	IV	Once daily	Renal function in obstructed labour
Metronidazole	500 mg	IV	8-hourly	Add if chorioamnionitis is suspected
Cefazolin	2 g	IV	Single prophylactic dose	At CS
Paracetamol	1 g	Oral	6-hourly	Perineal pain
Diclofenac	50 mg	Oral	8-hourly	Avoid if contraindicated (renal impairment, bleeding, asthma)
Vitamin K	1 mg	IM	Once	Newborn with subgaleal haemorrhage, if not already given

7. Guidelines, Policy and Law in Uganda

UGANDA CONTEXT

This topic follows the **Uganda Clinical Guidelines (UCG) 2023** (obstructed labour, perineal tears, PPH), the **Essential Maternal and Newborn Clinical Care Guidelines for Uganda**, the **WHO MCPC manual** (pudendal block, episiotomy repair, CS technique) and the **WHO 2018 intrapartum care recommendations**.

- **Partograph**: Ugandan labour wards use the **modified WHO partograph**, started in the active phase (cervix 4 cm or more). The **WHO 2018 recommendations** define the active first stage from **5 cm**, and the WHO **Labour Care Guide** (2020) replaces the partograph's alert and action lines with time limits per centimetre Verify *whether the Labour Care Guide has been adopted nationally in Uganda and which version is in current MoH use*. Know both: the Ugandan exam answer is the modified WHO partograph.
- **Episiotomy**: restrictive, not routine. WHO 2018 recommends **against routine or liberal episiotomy**; when it is needed, a **mediolateral** incision is used.
- **X-ray pelvimetry**: not recommended routinely; clinical pelvimetry and the progress of labour are the tests.
- **Obstructed labour**: managed at a CEmONC facility (HC IV and above). A HC III stabilises (IV fluids, catheter, antibiotics, **no oxytocin**) and refers (see the EmONC topic).
- **Obstetric fistula**: continuous bladder drainage for 10–14 days after prolonged obstructed labour, then referral to a fistula repair centre (e.g. Mulago, Kitovu Hospital, or regional fistula camps).
- **Female genital mutilation** is a criminal offence under the **Prohibition of Female Genital Mutilation Act, 2010**. A woman with FGM still needs respectful care and a delivery plan (an experienced attendant; anterior episiotomy or defibulation where needed) Verify *whether UCG 2023 gives a specific intrapartum FGM protocol*.
- **Consent and dignity**: every VE needs consent, privacy and a chaperone; the number of VEs should be kept to what labour monitoring needs (usually 4-hourly in the active phase), because repeated VEs raise the risk of infection.

> UGANDA CONTEXT
> **Where Ugandan practice and WHO differ**: active phase from **4 cm (modified WHO partograph, Uganda)** vs **5 cm (WHO 2018 / Labour Care Guide)**. Quote the Ugandan partograph in Ugandan exams and know the WHO change.

8. Key Points

High-yield points

- **In transverse, out AP**: the inlet is widest transversely (13 cm), the outlet widest AP (13 cm), the spines (10 cm) are the narrowest point and define station 0.
- **The head must present 9.5 cm** (well-flexed suboccipitobregmatic). Deflexion (11.5 cm) means a long labour; a brow (13–13.5 cm) cannot deliver at term.
- **Count the sutures**: 3 at the posterior fontanelle (Y), 4 at the anterior (diamond).
- **Believe the abdomen over the VE**: fifths palpable cannot be fooled by caput.
- **Moulding +3 with a high head is obstruction**: CS, not oxytocin or ventouse.
- **Water under the bridge**: the ureter passes under the uterine artery 1.5–2 cm lateral to the cervix.
- **Cardinal and uterosacral ligaments with levator ani** support the uterus; the round and broad ligaments do not.
- **Pain pathways**: first stage T10–L1; second stage pudendal S2–S4.
- **Catheter for 10–14 days** after obstructed labour prevents or heals early fistula.
- **A previous vaginal delivery of a good-sized baby** is the best pelvimeter.

Numbers & doses

Item	Value
Inlet diameters	AP (obstetric conjugate) **about 11 cm** (10.5–11); oblique **12 cm**; transverse **13 cm** (widest)
Diagonal conjugate	About **12.5 cm**; obstetric conjugate ≈ diagonal − **1.5 cm**

Item	Value
Interspinous diameter	About **10 cm**: the narrowest part; station 0
Outlet diameters	AP **13 cm** (coccyx back; widest); intertuberous **11 cm**
Subpubic angle (gynaecoid)	About **90°** (admits 2 fingers)
Suboccipitobregmatic (flexed vertex)	**9.5 cm**
Suboccipitofrontal (partly flexed)	**10 cm**
Occipitofrontal (deflexed / OP)	**11.5 cm**
Mentovertical (brow)	**13–13.5 cm**: no vaginal delivery at term
Submentobregmatic (face)	**9.5 cm**
Biparietal / bitemporal	**9.5 cm** / **8 cm**
Moulding reduces the engaging diameter by	Up to about **1 cm**
Anterior / posterior fontanelle closure	About **18 months** / about **6–8 weeks**
Engaged	Head **2/5 or less** palpable abdominally
Ventouse prerequisite	Head **1/5 or less** palpable; bony station at or below the spines
Uterine artery crosses the ureter	**1.5–2 cm lateral** to the cervix, at the internal os
Pudendal nerve	**S2–S4**; block just medial and posterior to the **ischial spine**
Labour pain pathways	First stage **T10–L1**; second stage **S2–S4**
Lidocaine without adrenaline (max)	**3 mg/kg** (about 200 mg)
Mediolateral episiotomy	About **60°** from the midline at crowning
Catheter after obstructed labour	**10–14 days**
Scar rupture risk	Lower segment about **0.5–1%**; classical about **4–9%**

Exam traps

- **"The pelvic inlet is widest AP."** → The inlet is widest **transversely** (13 cm); the **outlet** is widest AP. That is why the head enters OT and delivers OA.
- **"The fontanelle I feel is large, so it is the posterior one near the symphysis: OA."** → **Count the sutures**: 4 = anterior (diamond), 3 = posterior (Y). Feeling the anterior fontanelle easily means **deflexion/OP**, not OA.
- **"The head is at +2, so ventouse is fine."** → Check the **abdominal fifths**. A big caput can reach +2 while the bony head is still 3/5 palpable. Ventouse needs **1/5 or less** palpable.
- **"Poor progress, so start oxytocin."** → First exclude **obstruction** (moulding +3, high head, Bandl's ring, haematuria). Oxytocin into an obstructed uterus causes **rupture**.
- **"Brow presentation: augment and wait."** → Mentovertical 13–13.5 cm exceeds every pelvic diameter: **CS at term** unless it converts spontaneously.
- **"The round and broad ligaments hold the uterus up."** → The **cardinal and uterosacral ligaments** and **levator ani** are the supports.
- **"The ureter runs above the uterine artery."** → The ureter runs **under** it ("water under the bridge"), 1.5–2 cm lateral to the cervix.
- **"Cephalhaematoma crosses the sutures."** → **Caput** and **subgaleal haemorrhage** cross sutures; a cephalhaematoma is **limited by them**.

Questions seniors ask

Viva questions

1. **Why does the head enter the pelvis transversely and deliver OA?** The inlet's widest diameter is transverse (13 cm) and the outlet's is AP (13 cm). The gutter of the levator ani rotates the occiput forwards.
2. **Which diameter engages in a well-flexed vertex, and how long is it?** Suboccipitobregmatic, 9.5 cm (with the biparietal diameter of 9.5 cm transversely).
3. **Why can't a brow deliver vaginally at term?** The mentovertical diameter (13–13.5 cm) exceeds every pelvic diameter.
4. **What is the obstetric conjugate, and how do you estimate it?** Promontory to the nearest point on the back of the symphysis (about 11 cm): diagonal conjugate (about 12.5 cm) minus 1.5 cm.
5. **Where is the ureter at risk during CS or hysterectomy?** At the pelvic brim, near the infundibulopelvic ligament, and **under the uterine artery 1.5–2 cm lateral to the cervix**.
6. **What is the main support of the uterus?** The cardinal (Mackenrodt's) and uterosacral ligaments with levator ani.
7. **How do you tell caput from cephalhaematoma?** Caput crosses sutures and is present at birth. Cephalhaematoma is limited by the sutures and appears later.
8. **Nerve roots of the pudendal nerve?** S2, S3, S4 ("S2, 3, 4 keeps the pelvic floor off the floor").

On the ward round

How do you know the head is engaged? Abdominally, **2/5 or less** is palpable above the brim; vaginally, the bony leading part is at or near the spines. Engagement means the biparietal diameter (9.5 cm) has passed the inlet.

Show me how you do clinical pelvimetry. Promontory (not reached), sacral curve (hollow), side walls (parallel), spines (not prominent), sacrospinous ligament (2 fingers), subpubic angle (2 fingers, about 90°), intertuberous diameter (4 knuckles), coccyx (mobile).

On VE you feel a diamond-shaped fontanelle anteriorly. What is the position, and why does it matter? The anterior fontanelle anteriorly means **occipito-posterior** with **deflexion**: the occipitofrontal diameter (11.5 cm) presents, so expect a longer labour, backache and a higher chance of instrumental delivery or CS.

What are the signs of obstructed labour? Prolonged labour, **head 3/5 or more palpable** at full dilatation, **moulding +3**, large caput, oedematous cervix, **Bandl's ring**, haematuria, maternal tachycardia, fever and dehydration, and fetal distress.

Where can you injure the ureter at CS? Under the uterine artery 1.5–2 cm lateral to the cervix (lateral extensions, uterine artery ligation), at the **pelvic brim**, and near the **infundibulopelvic ligament**.

Why do we keep a catheter for 10–14 days after obstructed labour? Prolonged compression of the bladder base between head and pubis causes ischaemic necrosis. Continuous drainage rests the bladder and can let a small early fistula heal.

Why does the head rotate to OA? The occiput is the first part to meet the gutter-shaped **levator ani**, which guides it forwards; the **outlet is widest AP**, so the head's longest diameter aligns AP to exit.

Mnemonics

- **"In Transverse, Out Anteroposterior"**: the inlet is widest transversely and the outlet is widest AP.
- **Skull diameters "9.5, 10, 11.5, 13"**: **S**uboccipito**B**regmatic 9.5 (**S**mall and **B**est), suboccipito**F**rontal 10, **O**ccipito**F**rontal 11.5 (**O**P), **M**ento**V**ertical 13 (**M**ost **V**ast, brow).
- **Fontanelles**: **A**nterior is **A** bigger diamond (4 sutures); **P**osterior is **P**etite (3 sutures, Y).
- **"Water under the bridge"**: the ureter runs under the uterine artery.

- **"S2, 3, 4 keeps the pelvic floor off the floor"** (pudendal nerve, pelvic floor).
- **Ovarian veins: "Right goes Right to the IVC, Left takes a Left turn to the renal vein."**
- **Tube parts, medial to lateral: "In Is A Fine tube"** = **In**terstitial, **Is**thmus, **A**mpulla, **F**imbriae (with the infundibulum). Ectopics are commonest in the **A**mpulla.
- **Uterine supports: "a CUP holds the uterus"**: **C**ardinal, **U**terosacral, **P**ubocervical ligaments, sitting on the **P**elvic floor (levator ani). The round and broad ligaments are not supports.

OSCE tips

- On a **pelvis model**, point out the sacral promontory, ischial spines, sacrospinous ligament, subpubic arch and ischial tuberosities, and name the diameters of each plane.
- On a **fetal skull model**, show the fontanelles (count the sutures), demonstrate flexion versus deflexion, and name the presenting diameter for each attitude.
- **Demonstrate moulding grades** with your fingers on the model sutures.
- **Pelvimetry station**: describe it in the order inlet → cavity → mid-pelvis → outlet and finish with "clinically adequate" or "contracted at the … level" (Step 2).
- **VE report**: present it like the model VE report in section 4 (cervix, membranes and liquor, presenting part, position, station, moulding and caput, pelvis, impression).

> **EXAM FAVOURITE**
> Drawing a quick **table of skull diameters with presentations** and a **diagram of the pelvic inlet** earns marks in written papers. Practise until you can draw both in under 2 minutes.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Diameters of the pelvic inlet (AP, oblique, transverse)	**AP (obstetric conjugate) about 11 cm**, oblique about 12 cm, **transverse about 13 cm**. The widest diameter is transverse, so the head **enters in the occipito-transverse position**.
2	Diameters of the pelvic outlet	**AP about 13 cm** (coccyx displaced back), oblique about 12 cm, **transverse (intertuberous) about 11 cm**. The widest is AP, so the head **delivers OA**.
3	Narrowest part of the pelvis, and why it matters	The **mid-pelvis at the ischial spines**: **interspinous about 10 cm**. It is the reference point for **station (0)** and the level of internal rotation; prominent spines (android pelvis) cause **deep transverse arrest**.
4	Obstetric vs diagonal conjugate	**Obstetric conjugate:** promontory to the nearest point on the posterior symphysis, about 10.5–11 cm (the smallest AP the head must pass). **Diagonal conjugate:** promontory to the lower border of the symphysis, about 12.5 cm, measured on VE. Obstetric ≈ diagonal – 1.5 cm.
5	Four pelvic types (Caldwell–Moloy)	**Gynaecoid** (round, about 50%, ideal); **Android** (heart-shaped, prominent spines, narrow arch: deep transverse arrest, OP); **Anthropoid** (long AP oval: direct OP possible); **Platypelloid** (flat: transverse arrest at the inlet).
6	Fetal skull diameters: the 6 AP diameters	Suboccipitobregmatic **9.5** (flexed vertex); suboccipitofrontal **10**; occipitofrontal **11.5** (deflexed/OP); mentovertical **13–13.5** (brow); submentovertical **11.5**; submentobregmatic **9.5** (face).
7	Transverse diameters of the fetal skull	**Biparietal 9.5 cm** (the widest; passing the brim = engagement); **bitemporal 8 cm**; bimastoid 7.5 cm.
8	Anterior vs posterior fontanelle	**Anterior (bregma):** diamond, **4 sutures** (frontal, sagittal, 2 coronal), large, closes at about **18 months**. **Posterior (lambda):** triangle/Y, **3 sutures** (sagittal, 2 lambdoid), small, closes at **6–8 weeks**.
9	Grades of moulding	**0**: bones separated. **+1**: touching. **+2**: overlapping but reducible. **+3**: overlapping and **not reducible**, suggesting **CPD**.
10	Caput succedaneum vs cephalhaematoma vs subgaleal haemorrhage	**Caput:** scalp oedema, **crosses sutures**, present at birth, resolves in days. **Cephalhaematoma:** subperiosteal, **does not cross sutures**, appears after hours, jaundice. **Subgaleal:** under the aponeurosis, crosses sutures, fluctuant, **can cause shock** (vacuum).
11	Relationship of the uterine artery to the ureter	The uterine artery (anterior division of the internal iliac) crosses **above/anterior** to the ureter about **1.5–2 cm lateral to the cervix** at the level of the internal os: **'water under the bridge'**.
12	Main supports of the uterus	**Cardinal (transverse cervical/Mackenrodt's)** and **uterosacral** ligaments, **pubocervical** fascia, and the **levator ani** (pelvic floor). The round and broad ligaments are **not** true supports.
13	Pudendal nerve: roots, course, and where to block it	**S2–S4.** Leaves via the greater sciatic foramen, passes **behind the ischial spine/sacrospinous ligament**, and re-enters via the lesser sciatic foramen into **Alcock's canal**. Block: **just medial and posterior to the tip of the ischial spine**, aspirating first (internal pudendal vessels).
14	Nerve pathways of labour pain	**First stage** (uterus, cervix): visceral afferents to **T10–L1**. **Second stage** (vagina, perineum): **pudendal nerve S2–S4**.

#	Question	Answer
15	Ovarian blood supply and lymph drainage	**Ovarian artery from the aorta (L2).** Right ovarian vein to the **IVC**; left to the **left renal vein**. Lymph to the **para-aortic nodes**.
16	Parts of the fallopian tube and the commonest ectopic site	Interstitial, isthmus, **ampulla** (site of fertilisation and about 70–80% of ectopics), infundibulum with fimbriae. Total about 10 cm.
17	Lower vs upper uterine segment	**LUS:** from the isthmus (from about 28 weeks), thin, passive, loosely attached peritoneum; the **transverse CS** incision gives less bleeding and a stronger scar. **Upper segment:** thick, contracts and retracts; classical incision has a higher rupture risk.
18	Why do myometrial fibres stop bleeding after delivery?	The **middle myometrial layer** has interlacing (figure-of-eight) fibres around the vessels. When the uterus contracts they kink and compress the vessels: the **'living ligatures'**. Atony means they fail, causing PPH.
19	Lymph drainage of the vulva, cervix and uterine body	**Vulva/lower vagina:** superficial inguinal nodes. **Cervix:** parametrial, obturator, internal/external iliac nodes. **Uterine body:** iliac and para-aortic nodes (some to the inguinal nodes via the round ligament).
20	What is a Bandl's ring?	A **pathological retraction ring**: the rising junction between the thick retracting upper segment and the overstretched thin lower segment in **obstructed labour**. It warns of **impending uterine rupture**.
21	Clinical pelvimetry: the 8 things you assess, and the 'adequate' finding for each	**Promontory** not reached; **sacrum** hollow; **side walls** parallel; **spines** not prominent; **sacrospinous ligament** 2 fingers or more; **subpubic angle** 2 fingers (about 90°); **intertuberous diameter** 4 knuckles (about 8.5 cm or more); **coccyx** mobile.
22	VE says station +2, but the head is 3/5 palpable abdominally. Which do you believe?	**The abdomen.** A large **caput** can reach the spines while the bony head is still high. 3/5 palpable = **not engaged**. With +3 moulding this is **obstruction**: no ventouse (needs 1/5 or less palpable), no oxytocin; deliver by **CS**.
23	Signs of obstructed labour on the 'passage and passenger' workup	Head **3/5 or more palpable** at full dilatation, **moulding +3**, big caput, **oedematous, poorly applied cervix**, crossing the action line, **Bandl's ring**, haematuria, maternal tachycardia/dehydration/fever, fetal distress.
24	Deep transverse arrest vs obstruction: how do you tell them apart?	**DTA:** sagittal suture transverse with the head **engaged (1/5 or less palpable), at or below the spines, little moulding**: failure to rotate → ventouse/manual rotation by an experienced operator or CS. **Obstruction:** head **high (3/5 or more)**, **moulding +3**, caput: CS only.
25	What is the best evidence that a woman's pelvis is adequate?	A **previous vaginal delivery of a good-sized baby** (e.g. 3.5 kg or more). A primigravida's pelvis is 'untested'. **X-ray/CT pelvimetry is not recommended routinely**; labour itself (monitored on the partograph) is the real test.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 25-year-old primigravida is in labour at term. On VE the cervix is 9 cm dilated. You feel a large diamond-shaped depression with four sutures, supraorbital ridges and the root of the nose. The head is 3/5 palpable abdominally. What is the presenting diameter?

A Suboccipitobregmatic, 9.5 cm
B Occipitofrontal, 11.5 cm
C Submentobregmatic, 9.5 cm
D Mentovertical, 13–13.5 cm
E Biparietal, 9.5 cm

2. On VE in the active phase, you feel the sagittal suture in the right oblique diameter, with a small triangular fontanelle with three sutures anteriorly and to the left. What is the position?

A Right occipito-anterior (ROA)
B Left occipito-anterior (LOA)
C Left occipito-posterior (LOP)
D Right occipito-posterior (ROP)
E Direct occipito-anterior

3. During a caesarean hysterectomy for uncontrollable PPH at a HC IV, which step BEST prevents ureteric injury when securing the uterine vessels?

A Clamp the uterine vessels as laterally as possible at the pelvic side wall
B Ligate the internal iliac artery first in every case
C Avoid reflecting the bladder
D Clamp the infundibulopelvic ligament close to the pelvic brim without identifying the ureter
E Reflect the bladder down and clamp the uterine vessels close to the uterus at the level of the internal os

4. A 17-year-old primigravida, 145 cm tall, has been in labour for 18 hours at a HC III. The cervix is 7 cm with oedema, unchanged for 4 hours despite good contractions; there is moulding +3 and a large caput; the head is 3/5 palpable. The FHR is 150 bpm. What is the MOST appropriate management?

A Resuscitate (IV fluids, catheter, antibiotics) and refer urgently for caesarean section
B Start oxytocin augmentation
C Perform artificial rupture of membranes and reassess in 4 hours
D Vacuum delivery
E Encourage her to push

5. Which structure is the PRINCIPAL ligamentous support of the uterus, damage to which contributes to uterovaginal prolapse?

A Round ligament
B Broad ligament
C Cardinal (transverse cervical) ligament
D Ovarian ligament
E Infundibulopelvic ligament

6. You need to perform a pudendal block before an outlet vacuum delivery. Which landmark do you palpate vaginally to guide the needle?

A Sacral promontory
B Coccyx
C Ischial spine
D Ischial tuberosity
E Pubic symphysis

7. A newborn delivered by vacuum 6 hours ago has a boggy, fluctuant swelling extending from the forehead to the nape of the neck, crossing suture lines. HR 175 bpm, pale, capillary refill 4 seconds. What is the MOST likely diagnosis and priority?

A Caput succedaneum; reassure
B Cephalhaematoma; aspirate it
C Chignon; will resolve in hours
D Cephalhaematoma; phototherapy
E Subgaleal haemorrhage; urgent volume resuscitation and blood transfusion

8. In the second stage, the head has been at station +1 for 1 hour with the sagittal suture in the transverse diameter. Contractions are good. The pelvis has prominent ischial spines and a narrow subpubic arch. What is the diagnosis and the likely pelvic type?

A Occipito-posterior position; anthropoid pelvis
B Deep transverse arrest; android pelvis
C Brow presentation; gynaecoid pelvis
D Normal progress; gynaecoid pelvis
E Obstruction at the inlet; platypelloid pelvis

9. A 30-year-old woman has a left adnexal mass. The surgeon warns that the ovarian lymphatics drain to a distant group of nodes. Which nodes?

A Superficial inguinal
B Internal iliac
C Obturator
D Para-aortic nodes at the level of the renal vessels
E Sacral

10. A woman with a transverse lower segment CS scar asks why her risk of rupture in the next labour is lower than her friend's, who had a 'classical' CS. What is the BEST anatomical explanation?

A The lower segment is thin, passive and does not contract actively in labour, so its scar is under less strain and heals well
B The lower segment has more muscle, so it heals better
C The classical incision avoids the placental site
D The lower segment is supplied by the ovarian artery
E The upper segment has no peritoneal covering

11. A 28-year-old P3 is in labour at a HC IV. She had a normal vaginal delivery of a 3.6 kg baby 2 years ago. On VE the diagonal conjugate cannot be reached. The student wants to request X-ray pelvimetry. What is the MOST appropriate response?

A Agree; pelvimetry is needed for all multiparous women

B Explain that her previous vaginal delivery of a good-sized baby and the clinical findings suggest an adequate pelvis; X-ray pelvimetry is not routinely useful

C Perform an elective caesarean section

D Request a CT pelvimetry instead

E Measure the intertuberous diameter with an X-ray

12. After a 2-day obstructed labour ending in a CS for a stillbirth, a 19-year-old develops continuous leakage of urine from the vagina on day 8. What is the anatomical mechanism?

A Pudendal nerve damage causing stress incontinence

B Surgical ligation of the ureter

C Detrusor overactivity

D Prolonged compression of the bladder base and anterior vaginal wall between the fetal head and the pubic symphysis, causing ischaemic necrosis (vesicovaginal fistula)

E Rupture of the urethra during delivery

13. During a repair of a second-degree tear, which step is MOST important to prevent a missed anal sphincter injury and rectovaginal fistula?

A Using non-absorbable sutures

B Suturing the skin first

C Leaving the perineal muscles unsutured

D Inserting a vaginal pack for 48 hours

E Performing a rectal examination before and after the repair

14. A woman at 36 weeks has a head that is 2/5 palpable above the brim. What does this indicate?

A The head is engaged: the biparietal diameter has passed through the pelvic brim

B The head is not engaged

C The head is at station +3

D Cephalopelvic disproportion

E Placenta praevia

15. After a delivery with massive PPH, a surgeon at a regional referral hospital performs bilateral internal iliac (anterior division) ligation. The medical student asks why the uterus does not become necrotic. What is the BEST answer?

A The uterus is supplied only by the ovarian arteries

B The uterine artery arises from the external iliac artery

C Ligation reduces pulse pressure but collateral anastomoses (ovarian, lumbar, middle sacral, and others) maintain perfusion

D The ligation is always temporary

E The uterus does not need blood after delivery

Answers & explanations

1. D · Mentovertical, 13–13.5 cm

Anterior fontanelle + **supraorbital ridges + root of the nose** means a **brow presentation**, presenting the **mentovertical diameter (13–13.5 cm)**, the largest. It cannot pass a normal pelvis at term, so a CS is needed. A is a flexed vertex. B is deflexed/OP (no orbital ridges felt). C is a face presentation (mouth and malar bones felt). E is a transverse diameter, not the presenting AP diameter.

2. B · Left occipito-anterior (LOA)

The **posterior fontanelle** (small, 3 sutures, Y) marks the **occiput**. It is **anterior and on the mother's left**, so **LOA**, the commonest and most favourable position. A would have the occiput anterior-right. C and D would place the posterior fontanelle posteriorly. E would place the sagittal suture in the AP diameter.

3. E · Reflect the bladder down and clamp the uterine vessels close to the uterus at the level of the internal os

The ureter passes **under the uterine artery about 1.5–2 cm lateral to the cervix**. Pushing the **bladder** (and with it the ureters) down and clamping **close to the uterus** avoids it. A takes you towards the ureter. B is not routinely needed and has its own risks. C increases bladder and ureteric risk. D is a classic site of ureteric injury at the pelvic brim.

4. A · Resuscitate (IV fluids, catheter, antibiotics) and refer urgently for caesarean section

Short stature, adolescence, **+3 moulding**, **large caput**, **a high head (3/5)** and arrest despite **adequate contractions** mean **CPD/obstructed labour**. She needs resuscitation (fluids, **bladder catheter**, antibiotics) and **CS**. B: oxytocin in obstruction risks **uterine rupture**. C wastes time. D: vacuum needs full dilatation and an engaged head (2/5 or less). E: she is not fully dilated.

5. C · Cardinal (transverse cervical) ligament

The **cardinal ligaments** (with the uterosacral ligaments and levator ani) are the main supports. A: the round ligament maintains anteversion but gives little support. B: the broad ligament is a peritoneal fold. D and E suspend the ovary.

6. C · Ischial spine

The **pudendal nerve (S2–S4)** passes **behind the ischial spine** at the sacrospinous ligament. Inject just **medial and posterior to the tip of the spine**, after aspirating. The other landmarks are not related to the nerve's accessible course. The tuberosity (D) is used for a perineal (transperineal) approach, which is less reliable.

7. E · Subgaleal haemorrhage; urgent volume resuscitation and blood transfusion

A fluctuant swelling that **crosses sutures** and extends to the neck, after **vacuum**, with **tachycardia and poor perfusion**, is a **subgaleal haemorrhage**. The subaponeurotic space can hold much of a neonate's blood volume, so this is an emergency. A: caput is present at birth and does not cause shock. B and D: a cephalhaematoma is limited by the sutures, and it should never be aspirated. C: a chignon is localised to the cup site.

8. B · Deep transverse arrest; android pelvis

A head arrested **at or below the spines** with the sagittal suture **transverse** is **deep transverse arrest**. It is classically associated with the **android pelvis** (prominent spines, narrow arch). Management is rotation (manual or ventouse by an experienced operator) or CS. A: an anthropoid pelvis favours OA or direct OP. C: no brow findings. D: there is no progress. E: the head is below the inlet.

9. D · Para-aortic nodes at the level of the renal vessels

The ovaries descend from the posterior abdominal wall, so their vessels (the ovarian artery from the aorta at L2) and lymphatics run back up to the **para-aortic nodes**. A drains the vulva and lower vagina. B, C and E drain the cervix and upper vagina.

10. A · The lower segment is thin, passive and does not contract actively in labour, so its scar is under less strain and heals well

The **lower segment** (from the isthmus) is **thin and passive** and is not the contractile engine of labour. A transverse scar there has a rupture risk of about **0.5–1%**, compared with about **4–9%** for a **classical (upper segment) scar**, which is stretched by strong contractions. B: it has **less** muscle. C, D and E are false.

11. B · Explain that her previous vaginal delivery of a good-sized baby and the clinical findings suggest an adequate pelvis; X-ray pelvimetry is not routinely useful

The **best pelvimeter is a previous vaginal delivery** of a good-sized baby. Not reaching the promontory means a diagonal conjugate over about 12 cm. **Radiological pelvimetry does not improve outcomes** and is not recommended routinely. A, D and E add radiation without benefit. C is not indicated.

12. D · Prolonged compression of the bladder base and anterior vaginal wall between the fetal head and the pubic symphysis, causing ischaemic necrosis (vesicovaginal fistula)

Obstructed labour traps the **anterior vaginal wall and bladder base** between the fetal head and the **pubic bones**. Ischaemic necrosis sloughs at about **5–10 days**, causing a **vesicovaginal fistula** with continuous leakage. Prevention: **catheterise** in obstructed labour and keep the catheter for 10–14 days. A causes stress incontinence, not continuous leakage. B is possible after CS but is less likely than the classic pressure-necrosis story. C and E do not fit.

13. E · Performing a rectal examination before and after the repair

A **rectal examination before** the repair identifies 3rd/4th degree tears (the anal sphincter complex). One **after** the repair confirms that no suture has passed into the rectum (which can cause a fistula). A: use absorbable polyglactin. B: repair in layers from the vaginal apex down. C leaves the perineal body weak. D is not routine.

14. A · The head is engaged: the biparietal diameter has passed through the pelvic brim

Engagement means the **widest transverse diameter (biparietal, 9.5 cm)** has passed the brim, which corresponds to **2/5 or less palpable** abdominally. B: 3/5 or more is unengaged. C: station +3 is near the outlet (0/5). D and E suggest a **high** head, not an engaged one.

15. C · Ligation reduces pulse pressure but collateral anastomoses (ovarian, lumbar, middle sacral, and others) maintain perfusion

Internal iliac ligation **reduces pulse pressure** distal to the tie (turning arterial flow into something more like venous flow, which helps clotting), but rich **collaterals** (the ovarian artery anastomosis with the uterine artery, lumbar, iliolumbar and middle sacral arteries) keep the uterus alive, and future fertility has been reported. A and B are anatomically wrong. D: the ligatures are permanent. E is false.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The high head at full dilatation

Namutebi, an 18-year-old G1, 148 cm tall, is referred from a HC III to a HC IV after 20 hours of labour. The referral note says 'fully dilated for 2 hours, not delivering'. She is exhausted and has a temperature of 37.9 °C.

1. What will you look for on abdominal examination, and why?

> **Fifths of the head palpable** (3/5 or more at full dilatation means obstruction); a **Bandl's ring** and tense, tender lower segment (**impending rupture**); a distended bladder; contractions (frequency, strength); FHR; maternal pulse and signs of dehydration. Also check for a **visible bladder** and haematuria.

2. Abdomen: head 3/5 palpable, Bandl's ring at the umbilicus, FHR 172 bpm. VE: fully dilated but oedematous anterior lip, large caput reaching the spines, moulding +3, sagittal suture transverse. Interpret this.

> **Obstructed labour from CPD** (short adolescent pelvis) with **impending uterine rupture** (Bandl's ring) and **fetal distress**. The **caput** reaching the spines is **misleading**: the bony head is still high (3/5). She may also have intrapartum sepsis (fever).

3. What is your immediate management at the HC IV?

> Call the MO and theatre. **2 large-bore IV cannulae**, blood for Hb and **grouping and cross-match**, **IV fluids** (Ringer's lactate), **catheterise** (drain the bladder and monitor output), **antibiotics** (ampicillin 2 g IV + gentamicin 5 mg/kg IV + metronidazole 500 mg IV), **no oxytocin**, no vacuum, and move to theatre for an **emergency CS**.

4. What anatomical difficulties do you anticipate at CS, and how will you handle them?

> A **deeply impacted head** in the pelvis (an assistant pushes up from the vagina; or reverse breech extraction), a **thin, oedematous lower segment** at risk of lateral extensions into the **uterine vessels and broad ligament** (make a careful incision, curve it upwards, identify the **ureter** before suturing extensions), a bladder **pulled up** on the lower segment (reflect it carefully), and **PPH** from atony (oxytocin infusion, tranexamic acid).

5. What postoperative anatomy-related complication must you prevent, and how?

> **Obstetric fistula (VVF)** from pressure necrosis of the bladder base against the pubis: keep an **indwelling catheter for 10–14 days**, monitor for continuous leakage, and ensure good hydration. Also watch for **foot drop** (lumbosacral trunk compression), sepsis and PPH. Counsel her that she will need a **hospital birth (CS)** next time.

Take-away: Caput can fool your fingers: always correlate the VE station with the abdominal fifths. In obstructed labour, resuscitate, catheterise, give antibiotics and deliver by CS, and never give oxytocin.

Case 2: Reading the head on VE

Kyomugisha, a 27-year-old G2P1, is in labour at Mengo Hospital. Her first baby (3.4 kg) was delivered vaginally. At 4 hours after admission she is 8 cm dilated and complains of severe backache.

1. On VE you feel a large diamond-shaped fontanelle anteriorly on the right. What is the position, and what diameter is presenting?

> The **anterior fontanelle** (4 sutures, diamond) is anterior-right, so the **occiput is posterior-left: left occipito-posterior (LOP)**, with a **deflexed head** presenting the **occipitofrontal diameter (11.5 cm)** or suboccipitofrontal (10 cm). This explains the **backache** and slower progress.

2. What anatomical factors will help this labour?

> Her **previous 3.4 kg vaginal delivery** (proven pelvis), good contractions, **flexion** as the head meets the pelvic floor, and the **levator ani gutter** promoting **long anterior rotation** (about 3/8 of a circle) to OA. Most OP positions rotate spontaneously.

3. Two hours later she is fully dilated, the head is 1/5 palpable, at station +2, and still OP after 1 hour of pushing. The FHR is normal. What are the options?

> **Direct OP ('face to pubes') delivery** can occur (more perineal trauma, so consider an **episiotomy**); **manual rotation**; or **ventouse** (rotational with a posterior cup, e.g. Kiwi OmniCup) by a trained operator if criteria are met (fully dilated, head 1/5 or less, known position, ruptured membranes, adequate contractions, empty bladder). CS if the instrumental attempt fails or criteria are not met.

4. She delivers direct OP with a mediolateral episiotomy. How will you repair it?

> Do a **rectal exam first** to exclude an OASIS. **Lidocaine** top-up. **Polyglactin 2-0**: vaginal epithelium continuously from **1 cm above the apex** to the hymen; **perineal muscles** (bulbospongiosus, transverse perineal) to rebuild the **perineal body**; a **subcuticular** skin suture. Rectal exam after, and a swab and needle count.

5. What would you check in the newborn?

> Moulding and caput (expected), a **cephalhaematoma** or, if a ventouse was used, a **subgaleal haemorrhage** (check HR, pallor and a boggy swelling crossing sutures), and facial or scalp abrasions. Give vitamin K.

> **Take-away:** Count the sutures at the fontanelle to name the position. OP means a deflexed head with a bigger diameter and backache, and most OP positions rotate if powers and the pelvis are adequate.

Case 3: Bleeding at the corner of the CS incision

During an emergency CS at a regional referral hospital for obstructed labour, the lower segment incision extends laterally on the left, with brisk bleeding from the broad ligament. The intern is assisting you.

1. What are your first actions?

> **Apply pressure** with a pack, tell the anaesthetist (**blood loss, cross-match, fluids, tranexamic acid 1 g IV**), call for **senior help**, and improve exposure (extend the skin incision, retractors, suction, exteriorise the uterus if needed).

2. Which structures are at risk, and where exactly?

> The **uterine artery and vein** (running up the lateral uterine border from the base of the broad ligament) and the **ureter**, which lies **under the uterine artery about 1.5–2 cm lateral to the cervix** at the level of the internal os. The **bladder** is also at risk if the extension runs downwards.

3. How do you secure haemostasis safely?

> Identify the **apex of the tear**. Palpate the ureter (between thumb and finger at the base of the broad ligament, or trace it from the pelvic brim), then place sutures **close to the uterus**. If needed, do a **uterine artery ligation (O'Leary)**: a suture passed through the myometrium 2–3 cm medial to the vessels at the lower segment level. If bleeding persists, do an internal iliac ligation or hysterectomy by an experienced surgeon.

4. After surgery she has flank pain, fever and poor urine output. What do you suspect and how do you investigate?

> **Ureteric injury** (ligation or kinking). Check **creatinine** (it rises), do an **ultrasound** (hydronephrosis), and if available a **CT urogram** or IVU. Involve a urologist for stenting, deligation or reimplantation. Leakage of urine from the vagina days later suggests a **ureterovaginal fistula**.

5. How could this have been prevented?

> Make a **transverse incision curved upwards laterally**, and in obstructed labour use a gentle **blunt extension (fingers)** of the incision. Deliver an impacted head carefully (a vaginal push-up or reverse breech extraction) to avoid tearing, **reflect the bladder**, and have the anatomy of the ureter and uterine artery firmly in mind.

Take-away: 'Water under the bridge': when a lateral CS extension bleeds, pack, get help, and suture close to the uterus after locating the ureter.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.